A Formal Review of *Probabilistic Logic Networks* Ben Goertzel, Matthew Iklé, Izabela Freire Goertzel, Ari Heljakka (2009)

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Zar and Oruži[1]

[1]Oruži denotes AI-assisted collaboration using ChatGPT, Codex, and Claude Code.

Abstract

This document is a chapter-by-chapter formal review of the PLN book [1]. For each chapter, we evaluate what formalizes cleanly in Lean 4, what required correction, what fails outright (counterexamples), and what remains heuristic or operational.

This review is *book-relative*: it follows the PLN book's structure and uses the book as the reference against which the formalization is measured. For the self-contained mathematical refoundation, see *The World-Model Calculus* [2].

Every claim of "formalized" means: a Lean 4 theorem with zero `sorry`, building against Mathlib. Every claim of "corrected" means: the book's original statement is shown to be false in general, and a side-conditioned replacement is proved. Every claim of "counterexample" means: a concrete Lean witness that the stated property fails.

Contents

Chapter 1

How to Read This Review

1.1 Verdict Labels

Every theorem, formula, or claim from the PLN book is assigned one of five labels:

Formalized Cleanly The book's claim formalizes directly as stated, with at most minor notational changes.

Corrected Statement The book's claim is incorrect as stated but admits a corrected formulation with explicit side conditions. The original is false in general; the corrected version is proved.

Explicit Assumptions The claim holds under assumptions that the book left implicit. The formalization makes these assumptions explicit and proves the claim conditional on them.

Counterexample / No-Go A formal counterexample or no-go theorem showing that the book's claim fails in general. These are constructive Lean witnesses.

Heuristic / Operational The claim is operational, heuristic, or otherwise not amenable to theorem-level formalization. This includes runtime strategies, performance observations, and system-engineering advice.

1.2 What Counts as "Covered"

A PLN book chapter is *covered* when:

1. Its core mathematical claims have been formalized (possibly with corrections).
2. Its boundary conditions have been identified (counterexamples where claims fail).
3. Its operational/heuristic content has been explicitly marked as such.

A chapter is *not* covered merely by having been read or discussed. Coverage requires theorem-level Lean artifacts.

1.3 Relation to WM-PLN

Full theorem statements and proofs live in the companion volume, *The World-Model Calculus* [2]. This review provides:

- The book-relative context (what the original chapter was trying to do).
- The evaluation (what succeeded, what failed, what changed).
- Pointers to the corresponding WM-PLN chapters and Lean modules.

Chapter 2

Global Summary

Ch.	Topic	Verdict	Key finding
1	Introduction	Formalized Cleanly	Conceptual framing formalized as explicit semantic stack split.
2	Knowledge repr.	Formalized Cleanly	PLNQuery, WorldModel, WorldModelSigma capture the KR layer precisely.
3	Experiential sem.	Formalized Cleanly	Evidence-as-counts and revision algebra formalized exactly.
4	Indefinite TVs	Corrected Statement	Walley predictive and Bayesian credible intervals must be separated.
5	FO ext. inference	Corrected Statement	Classical formulas survive in corrected scope: BN exactness under screening-off/positivity, with no-go results showing that scope is necessary.
6	FO ext. + ITV	Formalized Cleanly	ITV transport through WM/OSLF bridge formalized.
7	Distributional TVs	Formalized Cleanly	Beta/Dirichlet layer, Kyburg flattening, de Finetti bridge.

8	Error magnification	Corrected Statement	Threshold transport needs explicit boundary conditions.
9	Large-scale inf.	Counterexample / No-Go	Multiple no-go results; corrected positives under explicit assumptions.
10	Higher-order ext.	Corrected Statement	Some unconditional equalities fail; side-conditioned replacements proved.
11	Quantifiers + ITV	Explicit Assumptions	Stable finite-domain bundle formalized; arbitrary-domain infinitary semantics already exist, but full parity is future work.
12	Intensional inf.	Corrected Statement	Settled in corrected typed three-channel form: ASSOC/PAT grounding, Solomonoff log-ratio bridge, selector endpoints, and explicit binary mixed-policy no-go.
13	Inference control	Formalized Cleanly	Selector specs, ranking, coverage formalized with canaries.
14	Temporal/causal	Formalized Cleanly	Event calculus grounding and temporal operators formalized.
App. A	PLN vs. NARS	Formalized Cleanly	Consolidated correspondence package with confidence transforms.

Chapter 3

Chapter 1: Introduction

3.1 What the Book Is Trying to Do

Chapter 1 introduces PLN's ambitions: a general-purpose uncertain reasoner combining logical structure with probabilistic semantics. It motivates truth values (strength and confidence), evidence-based reasoning, and the aspiration to handle deduction, induction, and abduction within a unified framework.

3.2 What Formalizes Cleanly

The conceptual framing—separating probabilistic semantics from evidence algebra from operational views—has been formalized as an explicit semantic stack split:

- **Layer 1** (world-model posterior): `Mettapedia/Logic/PLNWorldModel.lean`
- **Layer 2** (evidence algebra): `Mettapedia/Logic/EvidenceClass.lean`, `Mettapedia/Logic/EvidenceQuantale.lean`
- **Layer 3** (operational views): `Mettapedia/Logic/PLNIndefiniteTruth.lean`
- **Decision tree**: `Mettapedia/Logic/SemanticsDecisionTree.lean`

For the completed strength/confidence characterization itself, the current public endpoint is `Mettapedia/PLN/Core/PLNCanonicalAPI.lean` under `confidenceCharacterizationEndpointProfile`; the companion paper is `papers/pln-credal-degrees-of-freedom.tex`. This review points to that endpoint rather than re-deriving it here.

3.3 What Required Correction

The book does not explicitly separate the three layers, leading to the "semantic drift" discussed in WM-PLN Chapter 1. The correction is not a change in formulas but a change in *architecture*: making the layer boundaries explicit.

3.4 Counterexamples

None for this chapter. The introduction is interpretive, not formula-bearing.

3.5 What Remains Heuristic

The book's motivational examples and historical context are expository and not subject to formalization.

3.6 Verdict

FORMALIZED CLEANLY. The introduction's conceptual content formalizes well once the three-layer separation is made explicit. The refoundation is architecturally cleaner but preserves the original intuitions.

Chapter 4

Chapter 2: Knowledge Representation

4.1 What the Book Is Trying to Do

Chapter 2 defines PLN's knowledge representation: atoms, links, truth values, and the Atomspace data structure. It establishes how uncertain knowledge is stored and queried.

4.2 What Formalizes Cleanly

- `PLNQuery`: atoms, links, conditional links — `Mettapedia/Logic/PLNWorldModel.lean`
- `WorldModel`: the query interface — same file
- `WorldModelSigma`: side-condition-bundled models — same file
- Rewrite judgments: `Mettapedia/Logic/PLNWorldModelCalculus.lean`

4.3 What Required Correction

The book treats the Atomspace as a flat store of truth-valued links. The formalization adds:

- Typed query language separating prop/link/linkCond.
- Explicit side-condition contexts for structural assumptions.
- World-model revision as a first-class operation.

4.4 Counterexamples

None.

4.5 What Remains Heuristic

The Atomspace as a software system (indexing, storage, concurrency) is not formalized. The formalization captures the mathematical interface.

4.6 Verdict

Formalized Cleanly. The KR layer formalizes directly with natural type-theoretic representations.

Chapter 5

Chapter 3: Experiential Semantics

5.1 What the Book Is Trying to Do

Chapter 3 introduces the idea that PLN truth values are grounded in *evidence*: observations of positive and negative instances. The core representation is the evidence pair (n^+, n^-) where n^+ counts positive observations and n^- counts negative ones. Revision—combining two bodies of evidence—is componentwise addition:

$$(n_1^+, n_1^-) \oplus (n_2^+, n_2^-) = (n_1^+ + n_2^+,\ n_1^- + n_2^-)$$

The *strength* is the ratio $s = n^+/(n^+ + n^-)$ and the *confidence* is $c = n/(n+k)$ where $n = n^+ + n^-$.

The book's insight—that evidence counts are the right primitive, not point probabilities—is exactly right. De Finetti's theorem gives the mathematical justification: for exchangeable binary sequences, the counts are sufficient statistics.

5.2 What Formalizes Cleanly

- Evidence type (n^+, n^-) with componentwise operations — `Mettapedia/Logic/EvidenceClass.lean`
- Revision as additive monoid (associative, commutative, unit $(0, 0)$) — same file

- Information ordering (coordinatewise $\leq$) — same file
- Complete world-model posterior (Dirichlet pseudo-counts over 2^n worlds) — `Mettapedia/Logic/PLNJointEvidence.lean`
- Evidence compositionality: $\mathrm{ev}(W_1 + W_2, q) = \mathrm{ev}(W_1, q) \oplus \mathrm{ev}(W_2, q)$ — `Mettapedia/Logic/PLNWorldModel.lean`
- Fréchet bounds: $\max(0, s_A + s_B - 1) \leq s_{A \wedge B} \leq \min(s_A, s_B)$ — `Mettapedia/Logic/PLNFrechetBounds.lean`

5.3 What Required Correction

The book is essentially correct here. The formalization goes beyond the book by discovering additional algebraic structure:

- **Quantale structure:** a tensor product $\otimes$ on evidence that distributes over joins, capturing compositional dependent-evidence combination.
- **Heyting algebra:** a relative pseudo-complement $a \Rightarrow b$ satisfying $c \leq (a \Rightarrow b)$ iff $c \otimes a \leq b$. This means the evidence lattice supports intuitionistic logic.
- **Totality gate:** `Evidence` has incomparable elements (e.g., $(3, 1)$ and $(1, 3)$), so no faithful order-embedding into $\mathbb{R}$ exists. A single real number cannot capture all evidence information.

These are *additions*, not corrections. The book planted the seed; the formalization grew the full algebraic tree.

5.4 Counterexamples

- `threshold_or_counterexample`: because evidence values can be incomparable, disjunction threshold acceptance fails without assuming total-order comparability.

5.5 What Remains Heuristic

None. This is the most naturally formal chapter in the book.

5.6 Verdict

FORMALIZED CLEANLY. The evidence algebra is the strongest part of the original PLN book, and it formalizes beautifully. The formalization enriches it with quantale and Heyting structure that the book did not explore but that the evidence representation naturally supports.

Chapter 6

Chapter 4: Indefinite Truth Values

6.1 What the Book Is Trying to Do

Chapter 4 introduces indefinite truth values (ITVs): interval-valued representations of uncertainty that go beyond point estimates.

6.2 What Formalizes Cleanly

- ITV core with lower/upper/width/credibility coordinates — `Mettapedia/Logic/PLNIndefiniteTruth.lean`
- Walley IDM predictive interval constructors — same file, explicit `walleyIDM` constructors
- ITV hypercube transport — `Mettapedia/Logic/PLNWorldModelITVHypercube.lean`

6.3 What Required Correction

The book uses the notation $[L, U]$ without always specifying which interval semantics is intended. The formalization *insists* on explicit separation:

- **Bayesian credible intervals** (posterior-dependent, prior-sensitive).
- **Walley IDM predictive intervals** (prior-free, wider).

- **Typed selector choice**: the user declares which semantics.

This is a *strengthening*: the formalization is more precise than the book, not a rejection of the book's ideas.

6.4 Counterexamples

- `canary_ch11_rule4_not_equivalent_itvPath`: certain ITV transport paths are not equivalent, demonstrating that conflating interval semantics can produce different results.

6.5 What Remains Heuristic

The book's discussion of "intuitively reasonable" interval widths is expository rather than mathematical.

6.6 Verdict

Corrected Statement. The ITV idea was right; the formalization *strengthens* it by separating semantics that the book conflated.

Chapter 7

Chapter 5: First-Order Extensional Inference

7.1 What the Book Is Trying to Do

Chapter 5 presents the core PLN inference rules—deduction, induction, abduction, and revision—with strength formulas for computing truth values of derived links. The deduction formula, for instance, computes $P(C \mid A)$ from $P(B \mid A)$, $P(C \mid B)$, $P(B)$, and $P(C)$:

$$s_{AC} = s_{AB} \cdot s_{BC} + (1 - s_{AB}) \cdot \frac{s_C - s_B \cdot s_{BC}}{1 - s_B}$$

under the assumption that C is conditionally independent of A given B and given $\neg B$.

7.2 What Formalizes Cleanly

- Deduction strength formula: `plnDeductionStrength` — `Mettapedia/Logic/PLNDeduction.lean`
- Induction and abduction strength formulas — `Mettapedia/PLN/RuleFamilies/FirstOrder/PLNDerivation.lean`
- Full inference rule set — `Mettapedia/Logic/PLNInferenceRules.lean`
- MeTTa truth function parity / analysis — `Mettapedia/Logic/PeTTaLibPLNTruthFunctions.lean`, `Mettapedia/Logic/WMPLNJustifiedTruthFunctions.lean`, `Mettapedia/PLN/Comparisons/PeTTa/PeTTaLibPLNFormalAnalysis.lean`

- BN chain exactness: `chainBN_plnDeductionStrength_exact` — `Mettapedia/Logic/PLNBayesNetFastRules.lean`
- BN fork exactness: `forkBN_plnDeductionStrength_exact` — same file

The deduction formula is proved *exact* for the chain BN $A \to B \to C$ and the fork BN $A \leftarrow B \to C$. That is: under the respective d-separation conditions, the formula above equals $P(C \mid A)$ exactly—it is not an approximation.

7.3 What Required Correction

The book *does* mention conditional independence as a motivation for the deduction formula (Section 5.1 discusses the total probability decomposition, which implicitly assumes screening-off). However, the precise conditions under which each rule is exact are not always stated as formal hypotheses. The formalization's contribution is to recover the correct theoremic scope of the classical formulas by making these conditions fully explicit:

- Each rule carries a Σ-context stating the required conditional independences (e.g., $C \perp\!\!\!\perp A \mid B$ and $C \perp\!\!\!\perp A \mid \neg B$ for deduction).
- Positivity conditions ($s_B > 0$, $s_B < 1$) are stated as explicit hypotheses rather than tacit assumptions.
- In BN model classes, the needed side conditions are discharged by d-separation/local-Markov facts, so the rules are proved as *derived theorems* rather than treated as axioms or heuristics.

This is a corrected statement rather than an assumption patch. The classical extensional formulas survive, but only in the precise scope determined by screening-off, d-separation, and positivity, with exactness recovered on the corresponding BN-derived envelopes.

7.4 Counterexamples

- The local-completeness no-go theorem (WM-PLN Chapter 6) shows that no function of five local evidence values computes exact link evidence for *arbitrary* world models. This establishes that the screening-off conditions are not merely convenient—they are *necessary* for correctness.

This is different in kind from Chapter 11's current finite-domain front door: Chapter 5's side conditions are the mathematical content of the exact theorem, not a packaging gap between the reviewed chapter and a deeper semantic layer.

7.5 What Remains Heuristic

The choice of *when* to apply each rule (which links to try deduction on, in what order) is inference control, covered in Chapter 13.

7.6 Verdict

Corrected Statement. The book-era rule statements were too broad, but the classical extensional formulas survive in corrected form, with BN exactness theorems under explicit screening-off / d-separation / positivity conditions and no-go results showing that this scope is mathematically necessary.

Chapter 8

Chapter 6: First-Order Extensional Inference with ITVs

8.1 What the Book Is Trying to Do

Chapter 6 extends the first-order rules to operate on indefinite truth values (intervals) rather than simple truth values.

8.2 What Formalizes Cleanly

- ITV semantics transported through WM rewrites — `Mettapedia/PLN/Bridges/Languages/PLNWMOSLFBridgeITV.lean`
- One-call API bundles for ITV-coordinated inference — `Mettapedia/PLN/Core/PLNCanonicalAPI.lean`
- Coordinate-specialized coherence theorems — same file

8.3 What Required Correction

No fundamental correction needed. The formalization adds:

- Typed OSLF bridge with explicit selector specialization.
- Coordinate-by-coordinate transport (lower/upper/credibility/width/strength).

8.4 Counterexamples

None for this chapter specifically.

8.5 What Remains Heuristic

None.

8.6 Verdict

FORMALIZED CLEANLY. The ITV extension of first-order rules formalizes via the typed bridge infrastructure.

Chapter 9

Chapter 7: Distributional Truth Values

9.1 What the Book Is Trying to Do

Chapter 7 presents distributional truth values: truth values as entire distributions (Beta, Dirichlet) rather than point estimates or intervals. It introduces the Kyburg flattening operation.

9.2 What Formalizes Cleanly

- Context-aware strength as Beta posterior mean: `strengthWith_eq_beta_posterior_meanENN` — `Mettapedia/Logic/HigherOrder/PLNKyburgReduction.lean`
- Evidence aggregation as conjugate update: `hplus_is_bayesian_update` — same file
- Kyburg flattening: `kyburg_flattening`, `expectation_consistency`, `kyburg_no_advantage` — same file and `Mettapedia/ProbabilityTheory/HigherOrderProbability/KyburgFlattening.lean`
- Monad associativity for flattening: `flatten_associativity`, `flatten_associativity_kernel` — `Mettapedia/ProbabilityTheory/HigherOrderProbability/GiryMonad.lean`
- De Finetti singleton bridge into Kyburg flattening — `Mettapedia/Logic/HigherOrder/PLNKyburgReduction.lean`
- Worked example (uniform prior, evidence 3:1) — same file

9.2.1 Normal-Gamma Extension for Continuous Domains

The distributional story extends beyond Beta/Dirichlet to continuous observations via Normal-Gamma conjugate priors. Evidence becomes the sufficient statistic $(n, \sum x_i, \sum x_i^2)$; revision is again componentwise addition. The key compositionality theorem—batch replay equals sequential Bayesian update—is proven:

> `posterior_hplus_of_realizable`:
> $\text{posterior}(\pi, e_1 \oplus e_2) = \text{posterior}(\text{posterior}(\pi, e_1), e_2)$

See `Mettapedia/PLN/Bridges/ProbabilityTheory/EvidenceNormalGamma.lean`.

Six applied demos (broken sensor, widget factory, Kalman filter, gas sensor drift, steel faults, NAB anomaly) exercise these properties end-to-end on synthetic and real UCI/NAB data, totaling ~2,500 lines of Lean with 0 sorry; see the WM-PLN book, Ch. 7. A seventh script-level case study extends the same update semantics to the public Good Judgment Project / ACE forecasting stream: reliability-weighted aggregation improves substantially over mean/median pooling on held-out year-4 questions, and it also improves Brier/log score over stronger top-k log/logit pooling baselines while matching their accuracy. Empirically, the role split should be read as two flagships plus two supporting checks: GJP forecasting and Gas drift governance are the main applied stories, Steel is a supporting same-model-class abstention validation, and NAB is an honest negative boundary / case study. An explicit extremized-mean comparator is also weaker than the top-k logit family, so the WM gain is not just an artifact of beating plain averaging. The same WM-weighted pools also improve selective prediction AURC over those stronger baselines, while WM-on-top-k hybrids do not improve further, which is evidence that the gain comes from continuous reliability weighting rather than hard forecaster pruning. Nightly batch consolidation matches online updates to machine precision, and scoped forgetting of older years continues to improve recalibration even for the stronger pooling family. A small support-layer result also appears: on multi-option year-4 questions, gating the strong top-k log pool by WM agreement halves AURC from 0.001095 to 0.000541 while preserving accuracy, whereas the binary slice is an honest null because the strong top-k and WM-weighted pools already coincide everywhere. That forecasting study is not a new Lean theorem block, but it is a clean applied demonstration of the same conjugate evidence and revision laws.

9.3 What Required Correction

The Kyburg "no advantage" result is a sharpening: the book presents distributional truth values as potentially more powerful than simple truth values, but the formalization shows that for expectation-based decisions, they are equivalent (the flattening preserves expectation).

9.4 Counterexamples

None.

9.5 What Remains Heuristic

The book's discussion of when distributional truth values are "worth the computational cost" is a practical consideration, not a theorem.

9.6 Verdict

FORMALIZED CLEANLY. This is one of the strongest formalizations. The Kyburg flattening, de Finetti bridge, conjugate-update story, and Normal-Gamma continuous extension all formalize beautifully.

Chapter 10

Chapter 8: Error Magnification

10.1 What the Book Is Trying to Do

Chapter 8 discusses how errors accumulate through chains of inference ("error magnification") and proposes diagnostic approaches.

10.2 What Formalizes Cleanly

- View transport under WM query equivalence: strength, confidence, interpretation views all preserved — `Mettapedia/Logic/PLNWorldModelCalculus.lean`
- Confidence-threshold atom semantics and rewrite soundness — `Mettapedia/PLN/Bridges/Languages/PLNErrorMagnificationGrounding.lean`
- One-call WM rewrite → OSLF truth transport → threshold acceptance — `Mettapedia/PLN/Core/PLNCanonicalAPI.lean`
- WM consequence-rule layer with Kripke deontic lifts — `Mettapedia/PLN/Bridges/Logic/WorldModel/PLNWorldModelKripkeConsequence.lean`
- Categorical WM transport (institution, hyperdoctrine, Beck-Chevalley) — `Mettapedia/PLN/Bridges/CategoryTheory/WorldModel/PLNWorldModelCategoricalBridg lean`

- Kripke and neighborhood proof-theoretic closure (K, KT, KD, K4, E, EMN, EMT, ED) — `Mettapedia/PLN/Bridges/Logic/WorldModel/PLNWorldModelKripkeCompleteness.lean`, `Mettapedia/PLN/Bridges/Logic/WorldModel/PLNWorldModelNeighborhoodCompleteness.lean`

10.3 What Required Correction

The book's error magnification discussion is largely qualitative (graphs, chaos-plot intuitions). The formalization replaces this with:

- Precise threshold transport theorems with explicit boundary conditions.
- Double-damping strict gap characterization.
- Explicit counterexamples for disjunction and diamond threshold transport.

10.4 Counterexamples

- `threshold_or_counterexample`: disjunction threshold fails without total-order comparability.
- `threshold_dia_fails`: diamond threshold requires finite branching.
- `double_damping_strict_lt`: composed thresholds are strictly tighter than individual thresholds.

10.5 What Remains Heuristic

The book's chaos plots, sensitivity diagrams, and qualitative error-propagation discussions are expository. The formalized core is the threshold transport discipline with its precise boundaries.

10.6 Verdict

Corrected Statement. The book's qualitative discussion is replaced by precise transport theorems with explicit boundary conditions. The formalization is significantly more precise than the original.

Chapter 11

Chapter 9: Large-Scale Inference Strategies

This is the most complex chapter to review. We break it into subsections corresponding to the book's major topics.

11.1 What the Book Is Trying to Do

Chapter 9 covers strategies for scaling PLN to large knowledge bases: batch inference, multideduction, BN augmentation, trail-free protocols, and evidence pooling.

11.1.1 9.2: Batch Inference / BRGI

What formalizes.

- Finite BRGI object with policy-sensitive optimum characterization: `exists_optimal_feasible_set_with_policy_characterization` — `Mettapedia/PLN/InferenceControl/PremiseSelection/BRGI.lean`
- Graph-level BRGI with optimality-preserving refinement — same file

Verdict. Formalized Cleanly for the optimization-theoretic core. Heuristic / Operational for the runtime implementation.

11.1.2 9.3: Multideduction

The book proposes combining results from multiple deduction chains using inclusion-exclusion. The first two terms of IE give:

$$|A \cup B| \approx |A| + |B| - |A \cap B|$$

But the question is: can $|A \cup B|$ be determined from these two-term statistics alone?

What formalizes.

- **Identifiability no-go:** `same_first_two_terms_different_union` constructs two probability distributions agreeing on all first-two-term statistics but differing on $|A \cup B|$. Therefore two-term IE is *not identifiable*. — `Mettapedia/Logic/PLNInclusionExclusionIdentifiability.lean`
- **No universal predictor:** `no_universal_union_predictor_from_first_two_terms` — same file. No additive correction term works for all models.
- **Residual decomposition:** $|A \cup B| = T_2(A, B) + R(A, B)$ where T_2 is the two-term estimate and R is the residual. If R is known or bounded, the corrected estimate is exact: `correctedEstimate_agrees_of_residualAssumption` — `Mettapedia/Logic/PLNMultideductionResidual.lean`
- **N-way generalization:** the residual decomposition extends to n-way unions. — same file

Verdict. Counterexample / No-Go for two-term IE as an identifiable estimator. Explicit Assumptions for the corrected residual-based approach (requires knowing or bounding the residual R).

11.1.3 9.4: BN Augmentation

What formalizes.

- Class-packaged BN d-separation discharge — `Mettapedia/PLN/Bridges/ProbabilityTheory/BayesNet/PLNBNLocalMarkovPackages.lean`
- One-call selector → rewrite → threshold composition — `Mettapedia/PLN/Core/PLNCanonicalAPI.lean`
- Chain/fork/collider concrete endpoints — `Mettapedia/PLN/RuleFamilies/FirstOrder/PLNXiDerivedBNRules.lean`

Verdict. Formalized Cleanly. One of the strongest formalizations: exact BN query answering with class-packaged assumption discharge.

11.1.4 9.5: Trail-Free / OBI Protocols

The book proposes trail-free inference: updating truth values without tracking which inference paths contributed, to avoid exponential trail bookkeeping. The hope is that repeated application converges to a fixed point.

What formalizes.

- **Non-stabilization counterexample:** `orbit_not_eventually_constant` constructs a two-state system where the undamped trail-free update enters a 2-cycle $s_0 \to s_1 \to s_0 \to \cdots$ and never reaches a fixed point. The construction is explicit and decidable. — `Mettapedia/PLN/InferenceControl/ProtocolDynamics/PLNTrailFreeDynamicsCounterexample.lean`
- **Damped convergence:** with a damping factor $\alpha \in (0, 1)$ and eventual fresh evidence, the orbit eventually stabilizes: `damped_sp_spn_eventually_constant_of_ev` The inconsistency measure drops to zero. — `Mettapedia/PLN/InferenceControl/ProtocolDynamics/PLNTrailFreeDampedConvergence.lean`
- **Bounded-gap fairness:** if fresh evidence arrives with bounded gaps (at most g steps between arrivals), stabilization occurs within an explicit bound: `inconsistency_zero_within_bound_of_boundedGapFresh`. — same file
- **No-fresh negative:** without fresh evidence, even the damped protocol may not stabilize: `dampedOrbit_noFresh_not_eventually_constant`. — same file

Verdict. Counterexample / No-Go for undamped trail-free protocols. Explicit Assumptions for damped variants (requires fresh evidence with bounded gaps).

11.1.5 Evidence Pooling

What formalizes.

- Pooling axioms non-uniqueness: `maxPoolingOperator_ne_fuse` — `Mettapedia/PLN/InferenceControl/PremiseSelection/ExternalBayesianity.lean`
- Corrected uniqueness: external Bayes + total additivity — same file

Verdict. Counterexample / No-Go for pooling axioms alone. Explicit Assumptions with external Bayesianity.

11.1.6 Stochastic Experiment Layer

What formalizes.

- Blackwell factorization and expected utility — `Mettapedia/PLN/WorldModel/Experiment/PLNWorldModelExperimentStochastic.lean`
- Weighted source policies and trusted-gate transfer — same file

Verdict. Formalized Cleanly.

11.2 Overall Chapter 9 Verdict

Counterexample / No-Go for several core claims (identifiability, trail-free stabilization, pooling uniqueness). Explicit Assumptions for corrected positive replacements. Formalized Cleanly for the BN augmentation track. Operational runtime implementations remain Heuristic / Operational.

Chapter 12

Chapter 10: Higher-Order Extensional Inference

12.1 What the Book Is Trying to Do

Chapter 10 extends PLN to reason about relationships between predicates and concepts (higher-order extensional inference), including satisfying sets and reductions from higher-order to first-order.

12.2 What Formalizes Cleanly

- HOI-to-FOI reduction via satisfying sets: for predicate P, the satisfying set $\mathrm{Sat}(P) = \{x \mid P(x)\}$ bridges higher-order and first-order — `Mettapedia/Logic/HigherOrder/HigherOrderReduction.lean`
- Weakness-based inheritance: $\mathrm{Inheritance}(A, B, \mu) = \mathrm{weakness}(A \cap B)/\mathrm{weakness}(A)$ — `Mettapedia/Logic/HigherOrder/Basic.lean`
- Pointwise implication collapse: when $\forall x, R_1(A, x) \Rightarrow R_2(B, x)$ holds, inheritance equals 1 — same file
- Perfect inheritance corner ($\mathrm{pos} \neq 0$, $\mathrm{neg} = 0 \Rightarrow \mathrm{Inheritance} = \mathrm{pTrue}$) — same file

12.3 What Required Correction

The book's higher-order composition rule $\mathrm{Inh}(A, C) = f(\mathrm{Inh}(A, B), \mathrm{Inh}(B, C))$ is stated unconditionally. The formalization shows:

- The rule requires screening-off: $P(C \mid A, B) = P(C \mid B)$.
- Positivity conditions ($P(B) \in (0, 1)$) must hold.
- The corrected version carries these as explicit hypotheses, matching the structure of the first-order deduction correction.

12.4 Counterexamples

Where the book's unconditional equalities fail, the formalization provides concrete witnesses with specific evidence values showing the failure mode.

12.5 What Remains Heuristic

None of the core mathematical content.

12.6 Verdict

Corrected Statement. The reduction is sound under stated conditions; several book-era unconditional claims are shown to need screening-off and positivity side conditions, paralleling the first-order story.

Chapter 13

Chapter 11: Quantifiers with Indefinite Truth Values

13.1 What the Book Is Trying to Do

Chapter 11 extends PLN to quantified reasoning ("most birds fly," "few cats swim") using indefinite truth values for quantifier strength.

13.2 What Formalizes Cleanly

- Finite-model quantifier semantics with weakness connection — `Mettapedia/PLN/RuleFamilies/FirstOrder/Quantifiers/QuantifierSemantics.lean`
- Soundness theorems — `Mettapedia/PLN/RuleFamilies/FirstOrder/Quantifiers/Soundness.lean`
- Fuzzy quantifier semantics (QFM) — `Mettapedia/PLN/RuleFamilies/FirstOrder/Quantifiers/FuzzyQuantifierSemantics.lean`
- Fuzzy-to-ITV bridge — `Mettapedia/PLN/RuleFamilies/FirstOrder/Quantifiers/FuzzyITVBridge.lean`
- End-to-end canaries: `canary_ch11_rule_family_end_to_end_ext`, `fuzzyIntervalHolds_stren`
- Regression target: `Mettapedia/PLN/RuleFamilies/FirstOrder/Quantifiers/QuantifierRegression.lean`

13.3 What Required Correction

No fundamental corrections. The main clarification is scope: the reviewed chapter bundle is finite-domain, but an arbitrary-domain infinitary semantic layer already exists in Lean.

13.4 Counterexamples

- `canary_ch11_rule4_not_equivalent_itvPath`: certain ITV transport paths are not equivalent, demonstrating that some book-era assumptions about ITV composition are too strong.

13.5 What Remains Heuristic

Arbitrary-domain infinitary semantics already exist in `Mettapedia/PLN/RuleFamilies/FirstOrder/Quantifiers/Infinite.lean`; what remains beyond the current reviewed scope is stronger metatheory, chapter-level canary/regression parity, and measure-theoretic strengthening for probabilistic infinite spaces.

13.6 Verdict

Explicit Assumptions. Solid finite-domain theorem bundle, plus a real arbitrary-domain infinitary semantic layer that is not yet promoted to full chapter-level parity.

Chapter 14

Chapter 12: Intensional Inference

14.1 What the Book Is Trying to Do

Chapter 12 presents intensional inheritance: reasoning based on shared properties (intension) rather than shared instances (extension). It defines ASSOC and PAT channels and proposes mixing them with a parameter. The formal result is not verbatim preservation of that binary presentation, but a corrected typed three-channel account with an explicit no-go against collapsing back to the old binary law.

14.2 What Formalizes Cleanly

- Typed WM grounding for intensional inheritance — `Mettapedia/Logic/PLNIntensionalWorldModel.lean`
- Solomonoff-linked semantic transport — `Mettapedia/Logic/IntensionalInheritanceSolomono` `lean`
- Canonical `AssocPatSemanticModel` packaging — `Mettapedia/PLN/Core/PLNCanonicalAPI.lean`
- Selector-specialized end-to-end endpoints — same file
- Canary and regression targets — `Mettapedia/Logic/PLNIntensionalCanary.lean`, `Mettapedia/Logic/PLNIntensionalRegression.lean`

14.3 What Required Correction

The book proposes mixing extensional and intensional inheritance with a single parameter α:

$$\mathrm{InhInt}(A,B) = \alpha \cdot \mathrm{Ext}(A,B) + (1-\alpha) \cdot \mathrm{IntAssoc}(A,B)$$

The formalization shows this binary mixture is explicitly false as a general law in the corrected typed setting:

- The ASSOC channel uses mutual-information semantics: $P_{\mathrm{ASSOC}}(W \mid F) = P(W) \cdot 2^{I(F;W)}$.
- The PAT channel captures structural pattern proximity that mutual information misses.
- The concrete witness W_{demo} with scores $(3,2,1)$ for (ext, ASSOC, PAT) channels shows that the PAT channel contributes nontrivially and the mixed result cannot be expressed as any function of only the extensional and ASSOC values.

The corrected approach is not a patched binary rule. It is a typed three-channel model (extensional, ASSOC, PAT) with separate evidence extraction per channel, a semantic model that packages all three, and a constructive no-go theorem retiring the old binary law.

14.4 Counterexamples

- `binary_mixed_policy_collapse_no_go`: the old binary mixed law is explicitly false in the corrected typed setting. The constructive witness is `no_binary_mixed_policy_for_assocPat_fixture`, where pat_channel_nontrivial and mixed_not_assoc_only are both proved.

14.5 Claim-by-Claim Audit

- InheritanceSort = extensional | intensionalAssoc | intensionalPAT | mixed**, with separate channel projections.** PROVED. Endpoints: `InheritanceSort`, `InheritanceQueryFamily`, `extensionalEvidence`, `intensionalAssocEvidence`, `intensionalPATEvidence`, `mixedEvidence`.

- $P_{\text{ASSOC}}(W \mid F) = P(W) \cdot 2^{I(F;W)}$. Strengthened. The abstract information-theoretic formula is available as `goertzel_formula`; the corrected typed WM layer exports it through `AssocScoreCorrespondence` and `assocEvidence_eq_scoreToEvidence_of_assocScore_eq`.

- **PAT is a complementary structural channel, not reducible to ASSOC syntax alone.** Strengthened. Endpoints: `PATScoreCorrespondence`, `patEvidence_eq_scoreToEvidence_of_patScore_eq`, and `AssocPatSemanticModel`.

- **Universal-mixture intensional inheritance is the base-2 log ratio of conditionals.** Proved. Endpoints: `intensionalFromConditional_eq_log2_ratio`, `intensionalFromXiSemimeasure_eq_log2_ratio`, and `intensionalFromXiGeom_eq_log2_rati`

- **The chapter's Kolmogorov / normalized-information-distance gloss.** Strengthened. The exact formal endpoint is the universal-mixture log-ratio theorem above; the complexity-distance language is retained only as interpretation, not as the primary theorem statement.

- $\text{InhInt}(A, B) = \alpha \cdot \text{Ext}(A, B) + (1-\alpha) \cdot \text{IntAssoc}(A, B)$. Disproved. Endpoints: `binary_mixed_policy_collapse_no_go`, `mixed_not_assoc_only`, and `pat_channel_nontrivial`.

- **Concrete mixed projections and their divergence are explicit.** Proved. Endpoints: `canary_ch12_mixed_extensional_projection`, `canary_ch12_mixed_assoc_projection`, and `canary_ch12_mixed_projection_non_equivalen`

- **Typed ASSOC/PAT semantic packaging has selector-specialized end-to-end endpoints.** Proved. Endpoints: `mixedPolicy_assocPat`, `end_to_end_assocPat_bayesNormal_lower`, `end_to_end_assocPat_bayesExact_lower`, and `end_to_end_assocPat_walley_lower`.

14.6 Settlement

No chapter-level formula above remains open: each is now explicitly marked as proved, strengthened to the corrected typed statement, or ruled out. Any further alias polish would be presentational only and would not change the mathematical status of the chapter.

14.7 Verdict

Corrected Statement. The book's binary extensional+ASSOC mixture is too weak as stated. In the corrected typed setting, ASSOC and PAT live as separate channels with explicit semantic packaging, Solomonoff linkage, selector-specialized endpoints, and a constructive no-go against collapsing them to a binary mixed policy.

Chapter 15

Chapter 13: Aspects of Inference Control

15.1 What the Book Is Trying to Do

Chapter 13 discusses how to control PLN inference: which rules to apply, which premises to select, how to allocate computational resources.

15.2 What Formalizes Cleanly

- Formal selector specifications — `Mettapedia/PLN/InferenceControl/PremiseSelection/SelectorSpec.lean`
- Ranking transfer (optimality under distribution shift) — `Mettapedia/PLN/InferenceControl/PremiseSelection/Optimality.lean`
- Ranking stability — `Mettapedia/PLN/InferenceControl/PremiseSelection/RankingStability.lean`
- Coverage/submodularity with greedy guarantee — `Mettapedia/PLN/InferenceControl/PremiseSelection/Coverage.lean`
- Inference-control core composition — `Mettapedia/PLN/InferenceControl/PremiseSelection/PLNInferenceControlCore.lean`
- Canary and regression targets — `Mettapedia/PLN/InferenceControl/PremiseSelection/PLNInferenceControlCanary.lean`, `Mettapedia/PLN/InferenceControl/PremiseSelection/PLNInferenceControlRegression.lean`

15.3 What Required Correction

No fundamental corrections. The formalization adds:

- Submodularity (not in the book) with the $(1 - 1/e)$ greedy approximation guarantee.
- Coverage surrogate no-go: coverage and cardinality are insufficient for risk-sensitive selection.

15.4 Counterexamples

- `sameCoverageAndCard_differentFalsePositiveRisk`: coverage surrogate limitation.

15.5 What Remains Heuristic

The systems-engineering aspects of inference control (scheduling, resource allocation, attention) are operational and not theorem-level.

15.6 Verdict

Formalized Cleanly. The mathematical core of inference control formalizes well, with the submodularity story going beyond the book.

Chapter 16

Chapter 14: Temporal and Causal Inference

16.1 What the Book Is Trying to Do

Chapter 14 extends PLN to temporal reasoning (before/after/during) and causal inference (cause-effect relationships).

16.2 What Formalizes Cleanly

- Temporal predicate type: $\mathrm{TemporalPred}(D, T) = D \to T \to \mathrm{Prop}$
- Simultaneous operators: $\mathrm{SimAND}(P, Q)(x, t) = P(x, t) \wedge Q(x, t)$
- Sequential operators: $\mathrm{SeqAND}(P, Q)(x, t) = P(x, t) \wedge \exists \Delta > 0.\ Q(x, t + \Delta)$
- Predictive implication: $\mathrm{PredImpl}(P, Q)(x, t) = P(x, t) \Rightarrow \exists \Delta.\ Q(x, t + \Delta)$
- 3-node predictive chain with temporal bounds — `Mettapedia/PLN/RuleFamilies/Temporal/PLNTemporalCausalInference.lean`
- Probabilistic event calculus (Happens, Initiates, Terminates, HoldsAt) — `Mettapedia/PLN/RuleFamilies/Temporal/PLNProbabilisticEventCalculus.lean`
- One-call event endpoints — `Mettapedia/PLN/Core/PLNCanonicalAPI.lean`

16.3 What Required Correction

No corrections needed. The formalization treats temporal/causal inference as another world-model instance rather than a separate ontology. The key insight: temporal structure lives in the world-model state, temporal operators live in the query language, and the evidence/revision machinery operates unchanged.

16.4 Counterexamples

None.

16.5 What Remains Heuristic

The philosophical aspects of causation (interventionist vs. counterfactual vs. probabilistic) are not resolved by formalization; the formalization provides the probabilistic-event-calculus substrate. Granger-style temporal precedence is used to distinguish causation from correlation within the world-model framework, but this is a pragmatic rather than metaphysical choice.

16.6 Verdict

FORMALIZED CLEANLY. Compact core that formalizes as a WM instance. The temporal/causal layer is one of the cleanest demonstrations of the world-model approach: a single kernel, many instantiations.

Chapter 17

Appendix A: Comparison of PLN and NARS

17.1 What the Book Is Trying to Do

Appendix A compares PLN's inference rules with NARS (Non-Axiomatic Reasoning System), showing correspondences and differences.

17.2 What Formalizes Cleanly

- Consolidated PLN↔NARS comparison package — `Mettapedia/PLN/Comparisons/NARS/PLNNARSRuleCorrespondence.lean`
- Confidence transforms between PLN and NARS evidence — `Mettapedia/PLN/Comparisons/NARS/NARSEvidenceBridge.lean`
- Galois connection between evidence algebras — `Mettapedia/PLN/Comparisons/NARS/NARSPLNGaloisConnection.lean`

17.3 What Required Correction

None. The comparison is straightforward once both systems are formalized.

17.4 Counterexamples

None.

17.5 What Remains Heuristic

The qualitative comparison of "PLN philosophy vs. NARS philosophy" is not theorem-level.

17.6 Verdict

Formalized Cleanly. The rule correspondence and evidence bridge are fully formalized.

Appendix A

Full Theorem Ledger

The complete theorem ledger is maintained in the Lean codebase. Key entry points:

- `Mettapedia/PLN/Core/PLNCore.lean` — umbrella module
- `Mettapedia/PLN/Core/PLNCanonicalAPI.lean` — canonical API (3806 lines)
- `Mettapedia/Logic/README.md` — status table and theorem index
- `Mettapedia/Implementation/PLNParityChecklist.lean` — MeTTa parity

Chapter-specific regression scripts:

- `scripts/check\_ch8\_neighborhood.sh`
- `scripts/check\_ch9\_positive.sh`
- `scripts/check\_ch11\_quantifiers.sh`
- `scripts/check\_ch11\_fuzzy\_syllogism.sh`
- `scripts/check\_ch12\_intensional.sh`
- `scripts/check\_ch13\_inference\_control.sh`

Appendix B

Corrected Formulas

Book formula	Issue	Corrected version
Deduction: $s_{AC} = f(s_{AB}, s_{BC}, s_B, s_C)$ unconditionally	Missing screening-off / positivity scope	Same formula, but only under explicit screening-off and positivity side conditions
ITV: $[L, U]$ (unspecified semantics)	Conflates Bayesian credible and Walley predictive intervals	Explicit selector: Bayes-normal, Bayes-exact, or Walley-IDM
Intensional: $\text{InhInt} = \alpha \cdot \text{Ext} + (1 - \alpha) \cdot \text{Int}$	Binary mixed policy insufficient	Typed ASSOC/PAT channels with separate endpoints
Multideduction: two-term IE suffices	Identifiability no-go	Residual decomposition: $\|A \cup B\| = T_2 + R$
Trail-free: converges	2-cycle counterexample	Damped variant with fresh evidence
Pooling axioms: unique	Max-pooling satisfies axioms too	External Bayes + total additivity ⇒ unique

Appendix C

Counterexample Catalogue

Counterexample	**Module**	**What it blocks**
`same_first_two_terms_different_union`	PLNInclusionExclusionIdentifiability	Two-term IE identifying union mass
`orbit_not_eventually_constant`	PLNTrailFreeDynamicsCounterexample	Trail-free stabilization without damping
`maxPoolingOperator_ne_fuse`	PremiseSelectionExternalBayesianity	Pooling axioms characterizing additive fusion
`sameCoverageAndCard_differentFalsePositiveRisk`	PremiseSelectionCoverageCounterexamples	Coverage/cardinality surrogate for risk
`threshold_or_counterexample`	OSLFEvidenceSemantics	Disjunction threshold without total order
`threshold_dia_fails`	OSLFEvidenceSemantics	Diamond threshold without finite branching
`no_binary_mixed_policy_for_assocPat_fixture`	PLNIntensionalCanary	Collapsing ASSOC+PAT to binary law
No-go (local completeness)	PLNJointEvidenceNoGo	Computing exact link evidence from local premises

Appendix D

Deferred Claims

Claim	Why deferred	Status
Executable PLN engine	Requires MeTTa/Hyperon runtime	Heuristic / Operational
Arbitrary-domain quantifier parity	Basic infinitary semantics exist; full theorem/canary parity remains	Partial
Chapter 12 asserted claim set	No longer deferred: claim audit complete; chapter formulas are now explicitly proved, strengthened, or ruled out	Settled
Universal Prediction core	21-file WIP module	Not in stable narrative
Chapter 9 operational closure	Theorem-level complete; runtime pending	Heuristic / Operational

Appendix E

Notation Translation

PLN book	**Lean**	**WM-PLN notation**
$\langle s, c\rangle$	`STV s c`	$\langle s, c\rangle$
Strength s	`queryStrength`	$\mathsf{strength}(W, q)$
Confidence c	`queryConfidence`	$\mathsf{confidence}(W, q)$
Evidence (n^+, n^-)	`Evidence.mk nPos nNeg`	`Evidence`
Revision	`Evidence.add`	$e_1 \oplus e_2$
$P(B\|A)$	`link A B` query	$\mathtt{link}(A, B)$
Inheritance $A \Rightarrow B$	concept-side of `link`	Link query
Implication	predicate-side of `link`	Link query
World model	`WorldModel State Query`	`WorldModel`
Side conditions	`WorldModelSigma.sigma`	Σ
Indefinite TV $[L, U]$	`ITVSelector` choice	Typed ITV with selector

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